SAURABH RANJAN

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Ph.D. Cognitive Neuroscientist | Cognitive and Machine Psychology | Human-AI Alignment | NeuroAI

EDUCATION

Ph.D. Psychology — Behavioral & Cognitive Neuroscience University of Florida, USA | 2025

Dissertation: Reality Monitoring in Humans and Artificial Intelligence

M.Sc. Cognitive Science University of Allahabad, India | 2018

Thesis: Intentional Binding in Future-Directed Intentions

B.Sc. Physics Birla Institute of Technology, Mesra, India | 2016

TECHNICAL SKILLS

AI / ML & LLMs: evaluation pipelines for LLM safety and hallucination detection, agentic workflows, retrieval-augmented generation (RAG), prompt engineering; PyTorch, HuggingFace, LangChain, LangGraph, Pydantic, scikit-learn

Interpretability & Representational Analysis: linearizing encoding models and temporal generalization analysis (linear probing of internal representations across layers and time), psychological network analysis for comparing representational structure across human and model populations, multivariate signal decoding (MEG/EEG with MNE, fMRI with Nilearn/SPM), large-scale predictive modeling on UK Biobank RS-fMRI for phenotype prediction

Statistical Modeling & Evaluation Design: Bayesian hierarchical models (PyMC3, brms), signal detection theory, generalized mixed-effects models (lme4), multinomial processing trees, network analysis (bootnet, qgraph), causal & metacognitive modeling

Data Engineering & MLOps: Python, R, MATLAB, Bash; pandas, NumPy, Git/GitHub, SLURM/HPC, containerized environments, large-scale behavioral data collection (Prolific, Qualtrics, PsychoPy/JS)

Visualization & Communication: seaborn, matplotlib, plotly, ggplot2, tidyverse — publication-ready figures, dashboards, and stakeholder reporting

RESEARCH EXPERIENCE

Researcher — AI for Biomedical Health Outcomes *2026*

University of Florida, Gainesville, FL

- Build agentic AI systems (planning, reflection, tool use, multi-agent coordination) for clinical and medical decision-making, and develop instructional material that improves UF's AI readiness.
- Engineer reproducible research pipelines (PyTorch, LangChain/LangGraph, Pydantic, SLURM/HPC) that run and evaluate agentic workflows at scale.
- Co-develop UF's Master's-level curriculum on Data Science and Agentic LLMs for UF AI for Biomedical Health Sciences (AIBHS).

Graduate Researcher *2020 – 2025*

Odegaard Lab & UF Department of Psychology, Gainesville, FL

- Discovered that LLM source-attribution accuracy reverses under episodic memory delay, tested across 2 experiments and 6 transformer-based LLMs (Gemma3, Llama3.3, Llama4), a hallucination-relevant failure invisible to single-turn benchmarks.

- Found that corrective feedback produces two distinct metacognitive failure modes and that failure severity tracks active, not aggregate, parameter count, publishing the result as a preprint (arXiv:2607.23927).

- Designed and ran matched experiments with 100+ human participants and parallel LLM simulations, to generalize reality monitoring within human and LLM populations across task constraints.

- Applied neuroimaging decoding methods:
  - MEG temporal generalization to decode seen versus imagined experience
  - fMRI linearizing encoding models to predict fMRI response from image memorability in the NSD dataset

- Built psychological network models from 2,743 human participants and 6 LLMs (12B-272B parameters), showing human representational structure replicates across populations (r = 0.31-0.93) while LLMs fail to reproduce it at any scale, suggesting a world model can be characterized purely from the structural relationships between internally generated experiences in imagination and memory.

- Across these studies, LLMs showed distinct failures in knowing and accurately reporting their own knowledge, including at the largest scales tested (272B parameters) — a gap directly relevant to trusting and monitoring more capable future systems.

- Authored Psych Scanner, an open-source Python framework for running LLM cognitive experiments at scale.

- Mentored 7 undergraduate researchers and taught a full segment of Physiological Psychology.

Research Assistant *2019 – 2020*

Center of Behavioral and Cognitive Sciences, University of Allahabad, India

- Found stronger intentional binding, a measure of sense of agency, for predictive intermediate outcomes than non-predictive ones, replicated across three delays (300/500/700ms) and two contingency levels; published as an OSF preprint.

- This construct, monitoring one's own actions, now underlies how agentic AI systems track and self-correct across multi-step plans.

Research Assistant *2018*

Homi Bhabha Centre for Science Education, TIFR, Mumbai, India

- Contributed statistical analysis for Touchy-Feely Vectors (TFV), a gesture-based tool for teaching vectors, piloted across 266 grade-11 students (3 experimental, 3 control classrooms); the experimental group reasoned differently about vectors and showed higher engagement. Published in IEEE T4E 2019.

- Contributed statistical analysis for a 3-year field study of TFV (135 control vs. 131 experimental students), finding it improved model-based reasoning in stronger students and engagement in average ones. Published in the Journal of Computer Assisted Learning, 2021.

PUBLICATIONS

Working Papers

[1] **Ranjan, S.**, & Odegaard, B. Generalization of generation effect in reality monitoring.

Software

[2] **Ranjan, S.**, Makwana, M., Sokratous, K., & Odegaard, B. metasignal: A Python package for comprehensive metacognitive analysis and decision-making. Can be viewed as Preprint on Google Drive.

[1] **Ranjan, S.**, Sokratous, K., & Makwana, M. Psych Scanner: A Framework for Systematic Cognitive Evaluation of Large Language Models.

Blog

[1] **Ranjan, S.** (2026). "Why Time Feels the Way It Does." *Synthetic Selves* (Substack). https://syntheticselves.substack.com/p/why-time-feels-the-way-it-does

Theory and Empirical Peer-Reviewed & Preprints

[10] [Preprint] **Ranjan, S.,** Sokratous, K., & Odegaard, B. (2026). Reality Monitoring in Large Language Models: Self-Knowledge That Transforms with Conversation Memory. arXiv:2607.23927. https://arxiv.org/abs/2607.23927

[9] [Preprint] **Ranjan, S.**, & Odegaard, B. (2025). Psychological Imagination Networks Show Cross-Population Centrality and Clustering Alignment in Humans That Large Language Models Fail to Replicate. arXiv:2510.04391. https://arxiv.org/abs/2510.04391

[8] **Ranjan, S.**, & Odegaard, B. (2024). Heterarchy or hierarchy? Insights from a new model of visual imagination. *Physics of Life Reviews*, 49, 74–76.

[7] **Ranjan, S.**, & Odegaard, B. (2024). Reality monitoring and metacognitive judgments in a false-memory paradigm. *Neuroscience Research*, 201, 3–17.

[6] Maynes, R., Faulkner, R., Callahan, G., Mims, C. E., **Ranjan, S.**, Stalzer, J., & Odegaard, B. (2023). Metacognitive awareness in the sound-induced flash illusion. *Philosophical Transactions of the Royal Society B*, 378(1886), 20220347.

[5] Chiasson, P., Boylan, M. R., Elhamiasl, M., Pruitt, J. M., **Ranjan, S.**, Riels, K., … & Keil, A. (2023). Effects of neurofeedback training on performance in laboratory tasks: A systematic review. *International Journal of Psychophysiology*.

[4] Aggarwal, A., & **Ranjan, S.** (2022). How do undergraduate students reason about ethical and algorithmic decision-making? *Proc. 53rd ACM Technical Symposium on Computer Science Education*, 488–494.

[3] Karnam, D., Agrawal, H., Parte, P., **Ranjan, S.**, Borar, P., Kurup, P. P., … & Chandrasekharan, S. (2021). Touchy feely vectors: A compensatory design approach to support model-based reasoning. *Journal of Computer Assisted Learning*, 37(2), 446–474.

[2] Karnam, D., Agrawal, H., Parte, P., **Ranjan, S.**, Sule, A., & Chandrasekharan, S. (2019). Touchy feely affordances of digital technology for embodied interactions can enhance 'epistemic access'. *2019 IEEE Tenth International Conference on Technology for Education (T4E)*, 114–121.

[1] [Preprint] **Ranjan, S.**, & Srinivasan, N. (2019). Sense of agency for future-directed intentions. doi.org/10.31234/osf.io/qa93k.

INVITED TALKS

[2] Ranjan, S. (May, 2026). *The Spark of Artificial Neuroscience: Psychologically-Grounded Evaluation of Large Language Models.* Virtual invited talk, Autonomous Empirical Research Group + Laboratory for Automated Scientific Discovery of Mind and Brain (PI: Dr. Sebastian Musslick), Osnabrück University.

[1] Ranjan, S. (March, 2026). *Controlling Imagery Generation and its Awareness.* Virtual invited talk, Robert Reinhart Lab, Boston University.

SELECTED CONFERENCE POSTERS & TALKS

[8] **Ranjan, S.**, & Odegaard, B. (2025). Psychological Imagination Networks in Humans and LLMs. Frontiers in NeuroAI, Kempner Institute Symposium, Harvard University.

[7] **Ranjan, S.**, & Odegaard, B. (2025). Visual Imagination Networks in Humans and LLMs. Vision Sciences Society Annual Meeting, St. Pete, FL.

[6] **Ranjan, S.**, & Odegaard, B. (2024). The Fragility of Reality Monitoring under Extraneous Factors. Psychonomic Society 65th Annual Meeting, NYC.

[5] **Ranjan, S.**, & Odegaard, B. (2024). Reality Monitoring, Fast and Slow [Poster + Flash Talk]. Association for Psychological Science, San Francisco. *Awarded Scott O. Lilienfeld APS Travel Award.*

[4] Maw, M., Zhuang, L., Baltes, J., **Ranjan, S.**, & Odegaard, B. (2024). Task demands and sensory externalization in reality monitoring. PGSO Undergraduate Research Forum, UF. *Mentee won 3rd Prize.*

[3] Dundigalla, S., Johnson, D., Roh, A., **Ranjan, S.**, & Odegaard, B. (2024). Cognitive strategies in reality monitoring. PGSO Undergraduate Research Forum, UF.

[2] **Ranjan, S.**, Baltes, J., Roh, A., & Odegaard, B. (2023). Confidence in reality monitoring judgments. *Journal of Vision*, 23(9), 4844.

[1] **Ranjan, S.**, & Srinivasan, N. (2018). Intentional binding and future-directed intentions [Talk]. Annual Conference of Cognitive Science, IIT-Guwahati, India.

AWARDS & FELLOWSHIPS

- 2025 — College of Liberal Arts and Sciences Travel Award, University of Florida
- 2025 — Threadgill Dissertation Fellowship, University of Florida
- 2024 — Scott O. Lilienfeld APS Travel Award, Association for Psychological Science
- 2024–2025 — UF Department of Psychology Travel Awards (×4)
- 2020–2025 — Graduate Student Fellowship, UF Department of Psychology
- 2016–2018 — Graduate Merit Scholarship, Centre of Behavioral & Cognitive Sciences, University of Allahabad